

| Combination Numbers                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
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| <ul style="list-style-type: none"> <li>Pascal's Rule: <math>C_n^k = C_{n-1}^k + C_{n-1}^{k-1}</math>.</li> <li><b>Absorption Identity:</b> <math>kC_n^k = nC_{n-1}^{k-1}</math>.</li> <li>Trinomial Revision: <math>C_n^m C_m^k = C_n^{m-k} C_{n-k}^k</math>.</li> <li>Row Sum: <math>\sum_{k=0}^n C_n^k = 2^n</math>.</li> <li>Even-Odd Sums: <math>\sum_{k \geq 0} C_n^{2k} = \sum_{k \geq 0} C_n^{2k+1} = 2^{n-1}</math>.</li> <li>Weighted Sums: <ul style="list-style-type: none"> <li><math>\sum_{k=0}^n k C_n^k = n2^{n-1}</math>.</li> <li><math>\sum_{k=0}^n k^2 C_n^k = n(n+1)2^{n-2}</math>.</li> <li><math>\sum_{k=0}^n k^3 C_n^k = n^2(n+3)2^{n-3}</math>.</li> <li><math>\sum_{k=0}^m (-1)^k C_n^k = (-1)^m C_{n-1}^m</math>.</li> </ul> </li> <li>Hockey-Stick Identity: <math>\sum_{i=k}^n C_i^k = C_{n+1}^{k+1}</math>.</li> <li>Diagonal Sum: <math>\sum_{k=0}^m C_{n+k}^m = C_{n+m+1}^{m+1}</math>.</li> <li>Vandermonde's Identity: <math>\sum_{k=0}^r C_n^k C_{n-m}^{r-k} = C_{r+n}^r</math>.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Probability Space                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <ul style="list-style-type: none"> <li><b>Sample Space</b> <math>\Omega</math>: the set of all possible outcomes.</li> <li><b><math>\sigma</math>-algebra</b>: the collection of events. <ul style="list-style-type: none"> <li><math>\emptyset \in \mathcal{F}</math>,</li> <li>If <math>A \in \mathcal{F}</math>, then <math>A^0 \in \mathcal{F}</math>,</li> <li>If <math>A_i \in \mathcal{F}</math>, then <math>\bigcup_i A_i \in \mathcal{F}</math>.</li> </ul> </li> <li><b>Borel field</b>: all intervals and countable union of them.</li> <li><b>Probability Measure</b> <math>\mathbf{P}</math>: a function <math>\mathbf{P}: \mathcal{F} \rightarrow [0, 1]</math>. <ul style="list-style-type: none"> <li>Nonnegativity: <math>\mathbf{P}(A) \geq 0, \forall A \in \mathcal{F}</math>,</li> <li>Normalization: <math>\mathbf{P}(\emptyset) = 0, \mathbf{P}(\Omega) = 1</math>,</li> <li>Countable additivity: If <math>A_i \in \mathcal{F}</math> are <i>disjoint</i>, <math>\mathbf{P}(\bigcup_i A_i) = \sum_i \mathbf{P}(A_i)</math>.</li> </ul> </li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Conditioning and Independence                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <p><b>Conditioning</b></p> <ul style="list-style-type: none"> <li><b>Definition:</b> For <math>\mathbf{P}(B) \neq 0, \mathbf{P}(A B) = \mathbf{P}(A \cap B)/\mathbf{P}(B)</math>. <ul style="list-style-type: none"> <li>Nonnegativity: <math>\mathbf{P}(A B) \geq 0, \forall A \in \mathcal{F}</math>,</li> <li>Normalization: <math>\mathbf{P}(\Omega B) = 1</math>,</li> <li>Countable additivity: If <math>A_i \in \mathcal{F}</math> are <i>disjoint</i>, <math>\mathbf{P}(\bigcup_i A_i B) = \sum_i \mathbf{P}(A_i B)</math>.</li> </ul> </li> <li><b>Multiplication Rule:</b> <math>\mathbf{P}(A \cap B \cap C) = \mathbf{P}(A) \cdot \mathbf{P}(B A) \cdot \mathbf{P}(C A \cap B)</math>.</li> <li><b>Total Probability Theorem:</b> For a partition of sample space <math>A_1, A_2, \dots, A_n</math>, assuming <math>\mathbf{P}(A_i) &gt; 0</math> for all <math>i</math>,</li> </ul> $\mathbf{P}(B) = \sum_{i=1}^n \mathbf{P}(A_i) \mathbf{P}(B A_i).$ <ul style="list-style-type: none"> <li><b>Bayes's Rule:</b> <math>\mathbf{P}(A_i B) = \frac{\mathbf{P}(A_i) \mathbf{P}(B A_i)}{\sum_{j=1}^n \mathbf{P}(A_j) \mathbf{P}(B A_j)}</math>.</li> </ul> <p><b>Independence</b></p> <ul style="list-style-type: none"> <li><b>Definition:</b> Event <math>A</math> and <math>B</math> are called <b>independent</b> if <math>\mathbf{P}(A \cap B) = \mathbf{P}(A) \mathbf{P}(B)</math>.</li> <li>For <math>\mathbf{P}(B) &gt; 0, \mathbf{P}(A B) = \mathbf{P}(A)</math>.</li> <li><b>Conditional Independence:</b> <math>\mathbf{P}(A \cap B C) = \mathbf{P}(A C) \mathbf{P}(B C)</math>.</li> <li><b>Collective Independence:</b> Events <math>A_1, A_2, \dots, A_n</math> are called independent if <math display="block">\mathbf{P}(A_i \cap A_j \cap \dots \cap A_q) = \mathbf{P}(A_i) \mathbf{P}(A_j) \dots \mathbf{P}(A_q),</math> for any distinct indices <math>i, j, \dots, q</math> chosen from <math>\{1, 2, \dots, n\}</math>.</li> <li><b>Inclusion-Exclusion Principle:</b> <math>\mathbf{P}(\bigcup_{i=1}^n A_i) = \sum \mathbf{P}(A_i) - \sum_{i &lt; j} \mathbf{P}(A_i \cap A_j) + \dots + (-1)^{n-1} \mathbf{P}(A_1 \cap \dots \cap A_n)</math>.</li> <li><b>Hat Problem (Derangement):</b> <ul style="list-style-type: none"> <li>Derangement Number: <math display="block">D_n = n! \sum_{i=0}^n (-1)^i / i!</math> <math display="block">D_n = n D_{n-1} + (-1)^n.</math> <math display="block">\lim_{n \rightarrow \infty} D_n / n! = 1/e.</math> </li> <li>* Prob. of at least one match is <math>\approx 1 - 1/e</math> as <math>n \rightarrow \infty</math>.</li> </ul> </li> <li><b>Symmetric Difference:</b> <math>\mathbf{P}(A \triangle B) = \mathbf{P}(A) + \mathbf{P}(B) - 2\mathbf{P}(A \cap B)</math>.</li> </ul> |

| Discrete Random Variable                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
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| <ul style="list-style-type: none"> <li><b>Random Variable:</b> Given a probability space <math>(\Omega, \mathcal{F}, \mathbf{P})</math>, a <b>random variable</b> is a <i>function</i> <math>X: \Omega \rightarrow \mathbb{R}</math> such that <math>X^{-1}(A) \in \mathcal{F}</math> for each <math>A \in \mathcal{B}(\mathbb{R})</math>.</li> <li><b>PMF:</b> <math display="block">p_X(x) = \mathbf{P}(X = x).</math> </li> <li><b>Derived PMF:</b> <math>Y = g(X) \Rightarrow p_Y(y) = \sum_{x: g(x)=y} p_X(x)</math>.</li> <li><b>Expectation:</b> <math display="block">\mathbf{E}[X] = \sum_i x_i p_X(x_i).</math> </li> <li><b>Variance:</b> <math display="block">\text{var}[X] = \mathbf{E} \left[ (X - \mathbf{E}[X])^2 \right] = \mathbf{E}[X^2] - (\mathbf{E}[X])^2.</math> </li> <li><b><math>\mathbf{E}[\alpha X + \beta] = \alpha \mathbf{E}[X] + \beta, \mathbf{E}[X + Y] = \mathbf{E}[X] + \mathbf{E}[Y]</math>.</b></li> <li><math>\text{var}[\alpha X + \beta] = \alpha^2 \text{var}[X]</math>.</li> <li><b>Geometric RV:</b> <ul style="list-style-type: none"> <li><math>p_X(k) = (1 - p)^{k-1} p</math>,</li> <li><math>\mathbf{E}[X] = 1/p</math>,</li> <li><math>\mathbf{E}[X^2] = (2 - p)/p^2</math>,</li> <li><math>\text{var}[X] = (1 - p)/p^2</math>.</li> </ul> </li> <li><b>Memorylessness:</b> <math display="block">p_X(k X &gt; n) = (1 - p)^{k-n-1} p; \mathbf{E}[X X &gt; n] = 1/p + n.</math> </li> <li><b>Bernoulli RV:</b> <ul style="list-style-type: none"> <li><math>p_X(k) = \begin{cases} p, &amp; k = 1, \\ 1 - p, &amp; k = 0, \end{cases}</math></li> <li><math>\mathbf{E}[X] = p</math>,</li> <li><math>\mathbf{E}[X^2] = p</math>,</li> <li><math>\text{var}[X] = p(1 - p)</math>.</li> </ul> </li> <li><b>Discrete Uniform RV:</b> <ul style="list-style-type: none"> <li><math>p_X(k) = 1/(b - a + 1)</math>, if <math>k = a, a + 1, \dots, b</math>,</li> <li><math>\mathbf{E}[X] = (a + b)/2</math>,</li> <li><math>\text{var}[X] = (b - a)(b - a + 2)/12</math>.</li> </ul> </li> <li><b>Binomial RV:</b> <ul style="list-style-type: none"> <li><math>p_X(k) = C_n^k p^k (1 - p)^{n-k}</math>,</li> <li><math>\mathbf{E}[X] = np</math>,</li> <li><math>\mathbf{E}[X^2] = np(1 - p + np)</math>,</li> <li><math>\text{var}[X] = np(1 - p)</math>.</li> </ul> </li> <li><b>Poisson RV:</b> <b><math>\lambda = np</math> as <math>n \rightarrow \infty</math> in binomial RV.</b> <ul style="list-style-type: none"> <li><math>p_X(k) = e^{-\lambda} \lambda^k / k!, k = 0, 1, \dots</math>,</li> <li><math>\mathbf{E}[X] = \lambda</math>,</li> <li><math>\mathbf{E}[X^2] = \lambda + \lambda^2</math>,</li> <li><math>\text{var}[X] = \lambda</math>,</li> <li><b><math>X_1 + X_2 \sim \text{Poisson}(\lambda_1 + \lambda_2)</math>.</b></li> </ul> </li> </ul> |
| Multiple Discrete RV                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <ul style="list-style-type: none"> <li><b>Conditioning:</b> <ul style="list-style-type: none"> <li><b>Conditional PMF:</b> <math display="block">p_{X A}(x) = \mathbf{P}(\{X = x\} \cap A) / \mathbf{P}(A), p_{X Y}(x y) = p_{X,Y}(x, y) / p_Y(y).</math> </li> <li><b>Conditional Expectation:</b> <math display="block">\mathbf{E}[X A] = \sum_x x p_{X A}(x).</math> </li> <li><b>Total Expectation Theorem:</b> <math display="block">\mathbf{E}[X] = \mathbf{P}(A_1) \mathbf{E}[X A_1] + \dots + \mathbf{P}(A_n) \mathbf{E}[X A_n].</math> </li> <li><b>Marginal PMF:</b> <math display="block">p_{X,Y}(x, y) = \sum_z p_{X,Y,Z}(x, y, z); p_X(x) = \sum_y p_{X,Y}(x, y).</math> </li> </ul> </li> <li><b>Independence:</b> <ul style="list-style-type: none"> <li><math>\forall x, y: p_{X,Y}(x, y) = p_X(x) p_Y(y)</math>.</li> </ul> </li> <li><b>For Independent RVs:</b> <ul style="list-style-type: none"> <li><math>f(X)</math> and <math>g(Y)</math> are also independent.</li> <li><math>\mathbf{E}[XY] = \mathbf{E}[X] \mathbf{E}[Y]</math>.</li> <li><math>\text{var}(X + Y) = \text{var}(X) + \text{var}(Y), \text{var}(X - Y) = \text{var}(X) + \text{var}(Y)</math>.</li> <li><b>Convolution:</b> <math>W = X + Y \Rightarrow p_W(w) = \sum_x p_X(x) p_Y(w - x)</math>.</li> </ul> </li> <li><b>Conditional Independence:</b> <math display="block">\forall x, y: p_{X,Y}(x, y A) = p_X(x A) p_Y(y A).</math> </li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Continuous RV                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <ul style="list-style-type: none"> <li><b>PDF:</b> <math display="block">\mathbf{P}(a \leq X \leq b) = \int_a^b f_X(x) \, dx;</math> <math display="block">\mathbf{P}(x \leq X \leq x + \delta) \approx f_X(x) \delta.</math> </li> <li><b>Expectation:</b> <math display="block">\mathbf{E}[X] = \int_{-\infty}^{+\infty} x f_X(x) \, dx.</math> </li> <li><b>Variance:</b></li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

| $\text{var}(X) = \int_{-\infty}^{+\infty} (x - \mathbf{E}[X])^2 f_X(x) \, dx = \mathbf{E}[X^2] - (\mathbf{E}[X])^2.$ <ul style="list-style-type: none"> <li><b>Uniform RV:</b> <ul style="list-style-type: none"> <li><math>f_X(x) = 1/(b - a), a \leq x \leq b</math>,</li> <li><math>\mathbf{E}[X] = (a + b)/2</math>,</li> <li><math>\mathbf{E}[X^2] = (a^2 + ab + b^2)/3</math>,</li> <li><math>\text{var}[X] = (b - a)^2/12</math>.</li> </ul> </li> <li><b>Exponential RV:</b> <ul style="list-style-type: none"> <li><math>f_X(x) = \lambda e^{-\lambda x}, x \geq 0</math>,</li> <li><math>F_X(x) = 1 - e^{-\lambda x}, x \geq 0</math>.</li> <li><math>\mathbf{E}[X] = 1/\lambda</math>,</li> <li><math>\mathbf{E}[X^2] = 2/\lambda^2</math>,</li> <li><math>\text{var}[X] = 1/\lambda^2</math>.</li> <li><b><math>\mathbf{P}(X_1 &lt; X_2) = \lambda_1/(\lambda_1 + \lambda_2)</math>.</b></li> <li>Memorylessness: <math display="block">f_X(X &gt; a) = e^{-\lambda(X-a)}, \mathbf{P}(X \geq c + x X \geq c) = \mathbf{P}(X \geq x) = e^{-\lambda x};</math> <math display="block">\mathbf{E}[X X &gt; c] = 1/\lambda + c.</math> </li> </ul> </li> <li><b>Gaussian RV</b> (<math>X \sim N(\mu, \sigma^2)</math>): <ul style="list-style-type: none"> <li><math display="block">f_X(x) = \frac{1}{\sigma \sqrt{2\pi}} e^{-(x-\mu)^2/2\sigma^2},</math></li> <li><math>F_X(\lambda) = \Phi(\lambda), \Lambda \sim N(0, 1)</math>.</li> <li><math>\mathbf{E}[X] = \mu</math>,</li> <li><math>\mathbf{E}[X^2] = \mu^2 + \sigma^2</math>,</li> <li><math>\mathbf{E}[X^4] = 3\sigma^4 + 6\sigma^2 \mu^2 + \mu^4</math>,</li> <li><math>\text{var}[X] = \sigma^2</math>.</li> <li><math>Y = \alpha X + \beta \sim N(\alpha \mu + \beta, \alpha^2 \sigma^2)</math>.</li> </ul> </li> <li><b>CDF:</b> <ul style="list-style-type: none"> <li><math>F_X(x) = \int_{-\infty}^{x+} f_X(t) \, dt</math>.</li> <li><math>F_X(x)</math> discontinuous <math>\Rightarrow X</math> has discrete part.</li> <li><math>\mathbf{P}(a &lt; X \leq b) = F_X(b) - F_X(a)</math>.</li> <li><math>\mathbf{E}[X] = \int_0^{+\infty} (1 - F_X(x)) \, dx - \int_0^0 F_X(x) \, dx</math>.</li> </ul> </li> <li><b>Derived PDF:</b> <ul style="list-style-type: none"> <li><b>CDF Method:</b> find <math>F_Y(y) = \mathbf{P}(g(X) \leq y) \Rightarrow f_Y(y) = F_Y'(y)</math>.</li> <li><b>Linear:</b> <math>Y = \alpha X + b \Rightarrow f_Y(y) = \frac{1}{ \alpha } f_X\left(\frac{y-b}{\alpha}\right)</math>.</li> <li><b>Monotone:</b> <math>y = g(x) \Leftrightarrow x = h(y) \Rightarrow f_Y(y) = f_X(h(y)) h'(y) </math>.</li> </ul> </li> <li><b>Min and Max of i.i.d. RVs</b> <math>(X_1, \dots, X_n)</math>: <ul style="list-style-type: none"> <li><b>Max:</b> <math>Z = \max(X_i) \Rightarrow F_Z(z) = [F_X(z)]^n \implies f_Z(z) = n[F_X(z)]^{n-1} f_X(z)</math>.</li> <li><b>Min:</b> <math>W = \min(X_i) \implies F_W(w) = 1 - [1 - F_X(w)]^n</math>.</li> <li><i>Shortcut:</i> If <math>X_i \sim \text{Exp}(\lambda_i)</math>, then <math>\min(X_1, \dots, X_n) \sim \text{Exp}(\sum \lambda_i)</math>.</li> </ul> </li> <li><b>2D Uniform &amp; Geometric Probability:</b> <ul style="list-style-type: none"> <li><math>f_{X,Y}(x, y) = 1/\text{Area}(S)</math> for <math>(x, y) \in S</math>.</li> </ul> </li> <li><b>Broken Stick / Triangle Inequality:</b> Break a line into 3 pieces. They form a triangle iff. all pieces are shorter than 1/2. Prob. is 1/4.</li> </ul> |
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| Multiple Continuous RV                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <ul style="list-style-type: none"> <li><b>Conditioning:</b></li> <li><b>Conditional PDF:</b> <math>f_{X Y}(x y) = f_{X,Y}(x, y)/f_Y(y)</math>.</li> <li><b>Conditional Expectation:</b> <math>\mathbf{E}[X Y = y] = \int_{-\infty}^{+\infty} x f_{X Y}(x y) \, dx</math>.</li> <li><b>Total Expectation Theorem:</b> <math>\mathbf{E}[X] = \int_{-\infty}^{+\infty} \mathbf{E}[X Y = y] f_Y(y) \, dy</math>.</li> <li><b>Marginal PDF:</b> <math>f_X(x) = \int_{-\infty}^{+\infty} f_{X,Y}(x, y) \, dy</math>.</li> <li><b>Independence:</b> <ul style="list-style-type: none"> <li><math>\forall x, y: f_{X,Y}(x, y) = f_X(x) f_Y(y)</math>.</li> </ul> </li> <li><b>For independent RVs:</b> <ul style="list-style-type: none"> <li><math>f(X)</math> and <math>g(Y)</math> are also independent,</li> <li><math>\mathbf{E}[XY] = \mathbf{E}[X] \mathbf{E}[Y]</math>,</li> <li><math>\text{var}(X + Y) = \text{var}(X) + \text{var}(Y)</math>.</li> </ul> </li> <li><b>Convolution:</b> If <math>X</math> and <math>Y</math> are independent and <math>Z = X + Y, f_Z(z) = (f_X * f_Y)(z) = \int_{-\infty}^{+\infty} f_X(x) f_Y(z - x) \, dx</math>.</li> <li><math>X \sim N(\mu_X, \sigma_X^2), Y \sim N(\mu_Y, \sigma_Y^2) \Rightarrow X + Y \sim N(\mu_X + \mu_Y, \sigma_X^2 + \sigma_Y^2)</math>.</li> <li><b>Joint / Multiple CDF:</b> <math>F_{X,Y}(x, y) = \mathbf{P}(X \leq x, Y \leq y)</math>.</li> <li><b>Marginals:</b> <math>F_X(x) = F_{X,Y}(x, \infty), F_Y(y) = F_{X,Y}(\infty, y)</math>.</li> <li><b>From/To PDF:</b> <math display="block">F_{X,Y}(x, y) = \int_{-\infty}^x \int_{-\infty}^y f_{X,Y}(u, v) \, dv \, du</math> <math display="block">\implies f_{X,Y}(x, y) = \frac{\partial^2}{\partial x \partial y} F_{X,Y}(x, y).</math> </li> <li><b>Independence:</b> <math>X \perp Y \iff F_{X,Y}(x, y) = F_X(x) F_Y(y)</math>.</li> <li><b>Conditional CDF:</b></li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

| <ul style="list-style-type: none"> <li><b>Given Event</b> <math>A: F_{X A}(x) = \mathbf{P}(X \leq x \mid A) = \frac{\mathbf{P}(\{X \leq x\} \cap A)}{\mathbf{P}(A)}</math>.</li> <li><b>Given</b> <math>Y = y: F_{X Y}(x y) = \int_{-\infty}^x f_{X Y}(t y) \, dt</math>.</li> <li><b>Derivative:</b> <math>f_{X Y}(x y) = \frac{\partial}{\partial x} F_{X Y}(x y)</math>.</li> <li><b>Total Prob:</b> <math>F_X(x) = \sum \mathbf{P}(A_i) F_{X A_i}(x)</math>.</li> </ul> <p><b>Continuous Bayes' Rule:</b></p> $f_{X Y}(x y) = \frac{f_{Y X}(y x) f_X(x)}{f_Y(y)} = \frac{f_{Y X}(y x) f_X(x)}{\int_{-\infty}^{\infty} f_{Y X}(y t) f_X(t) dt},$ $\mathbf{P}(A Y = y) = \frac{\mathbf{P}(A) f_{Y A}(y)}{f_Y(y)}. (\text{Mixed Bayes})$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
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| Covariance and Correlation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <ul style="list-style-type: none"> <li><b>Covariance:</b> <math>\text{cov}(X, Y) = \mathbf{E}[(X - \mathbf{E}[X])(Y - \mathbf{E}[Y])] = \mathbf{E}[XY] - \mathbf{E}[X] \mathbf{E}[Y]</math>.</li> <li><b>Uncorrelated:</b> <math>\text{cov}(X, Y) = 0</math>; independent <math>\Rightarrow \text{cov}(X, Y) = 0</math>.</li> <li><math>\text{cov}(X, X) = \text{var}(X)</math>; <math>\text{cov}(X, \alpha Y + b) = \alpha \text{cov}(X, Y)</math>; <math>\text{cov}(X, Y + Z) = \text{cov}(X, Y) + \text{cov}(X, Z)</math>.</li> <li><math>\text{var}(X + Y) = \text{var}(X) + \text{var}(Y) + 2 \text{cov}(X, Y)</math>.</li> <li><math>\text{var}(\sum_i X_i) = \sum_i \text{var}(X_i) + \sum_{i \neq j} \text{cov}(X_i, X_j)</math>.</li> <li><b>Correlation:</b> <math>\rho(X, Y) = \frac{\text{cov}(X, Y)}{\sqrt{\text{var}(X) \text{var}(Y)}}</math>, <math>-1 \leq \rho \leq 1</math>.</li> <li><math> \rho  = 1 \Leftrightarrow X - \mathbf{E}[X] = c(Y - \mathbf{E}[Y])</math>.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Conditional Expectation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <ul style="list-style-type: none"> <li><b>Iterated Expectation:</b> <math>\mathbf{E}[X] = \mathbf{E}[\mathbf{E}[X Y]]</math>. <ul style="list-style-type: none"> <li><math>Y</math> discrete: <math>\mathbf{E}[\mathbf{E}[X Y]] = \sum_y \mathbf{E}[X Y = y] p_Y(y)</math>.</li> <li><math>Y</math> continuous: <math>\mathbf{E}[\mathbf{E}[X Y]] = \int \mathbf{E}[X Y = y] f_Y(y) \, dy</math>.</li> </ul> </li> <li><b>Estimator:</b> <math>\hat{X} = \mathbf{E}[X Y], \tilde{X} = X - \hat{X}</math>.</li> <li><math>\mathbf{E}[\tilde{X} Y] = 0 \Rightarrow \mathbf{E}[\tilde{X}] = 0; \text{cov}(\tilde{X}, \hat{X}) = 0</math>.</li> <li><b>Conditional Variance:</b> <math>\text{var}(X Y) = \mathbf{E}[(X - \mathbf{E}[X Y])^2 Y]</math>.</li> <li><b>Total Variance:</b> <math>\text{var}(X) = \text{var}(\mathbf{E}[X Y]) + \mathbf{E}[\text{var}(X Y)]</math>.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Transforms                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <ul style="list-style-type: none"> <li><b>Definition:</b> <math>\mathcal{M}_X(s) = \mathbf{E}[e^{sX}]</math>; discrete <math>\sum_i p_X(x_i) e^{sx_i}</math>; continuous <math>\int f_X(x) e^{sx} \, dx</math>.</li> <li><math>\mathcal{M}_X(0) = 1; \mathbf{E}[X^n] = \left. \frac{d^n \mathcal{M}_X(s)}{ds^n} \right _{s=0}</math>.</li> <li><math>Y = \alpha X + b \Rightarrow \mathcal{M}_Y(s) = e^{bs} \mathcal{M}_X(as)</math>.</li> <li>Independent <math>Z = \sum_{i=1}^n X_i \Rightarrow \mathcal{M}_Z(s) = \prod_i \mathcal{M}_{X_i}(s)</math>.</li> <li><b>Mixture:</b> <math>f_Y = \sum_i p_i f_{X_i} \Rightarrow \mathcal{M}_Y(s) = \sum_i p_i \mathcal{M}_{X_i}(s)</math>.</li> <li><b>Random Sum:</b> <math>Y = X_1 + \dots + X_N</math>, i.i.d. <math>X_i \perp N</math>: <ul style="list-style-type: none"> <li><math>\mathbf{E}[Y] = \mathbf{E}[N] \mathbf{E}[X]</math>,</li> <li><math>\text{var}(Y) = \mathbf{E}[N] \text{var}(X) + (\mathbf{E}[X])^2 \text{var}(N)</math>,</li> <li><math>\mathcal{M}_Y(s) = \mathcal{M}_N(\ln \mathcal{M}_X(s))</math>.</li> </ul> </li> <li><b>Common <math>\mathcal{M}_X(s)</math>:</b> <ul style="list-style-type: none"> <li>Bernoulli: <math>1 - p + pe^s</math>; Binomial: <math>(1 - p + pe^s)^n</math>,</li> <li>Geometric: <math>\frac{pe^s}{1 - (1 - p)e^s}</math>; Poisson: <math>\exp(\lambda(e^s - 1))</math>,</li> <li>Discrete uniform <math>\{a, \dots, b\}</math>: <math>\frac{e^{as}(1 - e^{(b-a+1)s})}{(b - a + 1)(1 - e^s)}</math>,</li> <li>Uniform <math>[a, b]</math>: <math>\frac{e^{bs} - e^{as}}{(b - a)s}</math>; Exponential: <math>\frac{\lambda}{\lambda - s}</math>,</li> <li>Gaussian <math>N(\mu, \sigma^2)</math>: <math>\exp(\mu s + \sigma^2 s^2/2)</math>.</li> </ul> </li> </ul> |
| Probability Inequalities                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <ul style="list-style-type: none"> <li><b>Markov:</b> <math>X \geq 0, a &gt; 0 \Rightarrow \mathbf{P}(X \geq a) \leq \mathbf{E}[X]/a</math>. <ul style="list-style-type: none"> <li>"=" iff <math>X \in \{0, a\}</math>.</li> </ul> </li> <li><b>Chebyshev:</b> <math>\mathbf{P}( X - \mathbf{E}[X]  \geq c) \leq \text{var}(X)/c^2</math>. <ul style="list-style-type: none"> <li>"=" iff <math>X \in \{\mu - c, \mu, \mu + c\}</math>, <math>\mathbf{P}(X = \mu \pm c) = \sigma^2/(2c^2)</math>.</li> </ul> </li> <li><b>Variance Bound:</b> <math>X \in [a, b] \Rightarrow \text{var}(X) \leq (b - a)^2/4</math>. <ul style="list-style-type: none"> <li>"=" iff <math>\mathbf{P}(X = a) = \mathbf{P}(X = b) = 1/2</math>.</li> </ul> </li> <li><b>Chernoff:</b> <ul style="list-style-type: none"> <li>Right-tail: <math>\mathbf{P}(X \geq a) \leq \inf_{s \geq 0} e^{-sa} \mathcal{M}_X(s) = e^{-\sup_{s \geq 0} (sa - \ln \mathcal{M}_X(s))}</math>,</li> <li>Left-tail: <math>\mathbf{P}(X \leq a) \leq \inf_{s \leq 0} e^{-sa} \mathcal{M}_X(s) = e^{-\sup_{s \leq 0} (sa - \ln \mathcal{M}_X(s))}</math>.</li> </ul> </li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

LLN and CLT

- **WLLN:**  $M_n = \frac{1}{n} \sum_{i=1}^n X_i$ , i.i.d.,  $E[X_i] = \mu$ ,  $\text{var}(X_i) = \sigma^2$ .
  - $E[M_n] = \mu$ ,  $\text{var}(M_n) = \sigma^2/n$ ,
  - $P(|M_n - \mu| \geq \epsilon) \leq \sigma^2/(n\epsilon^2) \rightarrow 0$ ,
  - $M_n \xrightarrow{P} \mu$ .
- **Convergence:**
  - **In Probability:**  $X_n \xrightarrow{P} X \Leftrightarrow P(|X_n - X| \geq \epsilon) \rightarrow 0, \forall \epsilon > 0$ ,
  - **In Distribution:**  $X_n \xrightarrow{D} X \Leftrightarrow F_{X_n}(x) \rightarrow F_X(x)$  at continuous  $x$ ,
  - **Almost Surely:**  $X_n \xrightarrow{\text{a.s.}} X \Leftrightarrow P(\lim_n X_n = X) = 1$ ,  
\* "Almost Surely": For almost all sample paths  $\omega$ .
  - **$r$ -th Mean:**  $X_n \xrightarrow{L^r} X \Leftrightarrow E[|X_n - X|^r] \rightarrow 0$ ,
  - **Relations:** a.s.  $\Rightarrow P \Rightarrow D$ ;  $r$ -mean  $\Rightarrow P \Rightarrow D$ ;  $r \geq s \Rightarrow L^r \Rightarrow L^s$ .
- **CLT:**  $S_n = \sum_{i=1}^n X_i$ , i.i.d.,  $E[X_i] = \mu$ ,  $\text{var}(X_i) = \sigma^2$ .
  - $Z_n = \frac{S_n - n\mu}{\sigma\sqrt{n}} \xrightarrow{D} N(0, 1)$ ;  $P(Z_n \leq z) \rightarrow \Phi(z)$ .
- **Transform:**  $\mathcal{M}_{Z_n}(s) = e^{-s\mu\sqrt{n}/\sigma} \left( \mathcal{M}_{X_1} \left( \frac{s}{\sigma\sqrt{n}} \right) \right)^n \rightarrow e^{s^2/2}$ .
- **Normal Approx.:**  $S_n \approx N(n\mu, n\sigma^2)$ ;  $P(S_n \leq c) \approx \Phi\left(\frac{c - n\mu}{\sigma\sqrt{n}}\right)$ .
- **Binomial Approx.:**  $X_i \sim \text{Bernoulli}(p)$ ,  $S_n = \sum_i X_i \sim \text{Bin}(n, p)$ ,  $q = 1 - p$ .
  - $\frac{S_n - np}{\sqrt{npq}} \approx N(0, 1)$ ,
  - $P(k \leq S_n \leq l) \approx \Phi\left(\frac{l + 0.5 - np}{\sqrt{npq}}\right) - \Phi\left(\frac{k - 0.5 - np}{\sqrt{npq}}\right)$ .
- **SLLN:**  $M_n = S_n/n \Rightarrow P(\lim_{n \rightarrow \infty} M_n = \mu) = 1$ , i.e.  $M_n \xrightarrow{\text{a.s.}} \mu$ .

Event Limits and Borel-Cantelli

- **Set Limits:**
  - $\limsup A_n = \bigcap_{k \geq 1} \bigcup_{n \geq k} A_n = \{A_n \text{ i.o.}\}$ ;  
\* "infinite often" =  $\forall N \geq 0, \exists n \geq N, A_n$  happens.
  - $\liminf A_n = \bigcup_{k \geq 1} \bigcap_{n \geq k} A_n = \{A_n \text{ eventually}\}$ .  
\* "eventually" =  $\exists N \geq 0, \forall n \geq N, A_n$  happens.
  - $\liminf A_n \subseteq \limsup A_n$ ; " = "  $\Rightarrow \lim A_n$  exists.
- **Monotone Convergence Theorem:**  $0 \leq X_n \uparrow X \Rightarrow E[X_n] \uparrow E[X]$ ; events  $A_n \uparrow A \Rightarrow P(A_n) \uparrow P(A)$ .
  - Let  $X_n$  = the sum of an RV seq.  $\{Y_i\}$ ,  $Y_i \geq 0$ ,  $X = \lim_n X_n$ ,  
 $E[X] < \infty \Rightarrow P(X = \infty) = 0 \Rightarrow X$  converge a.s..
  - $X = \sum_{i=1}^{\infty} Y_i$  converges a.s.  $\Rightarrow \{Y_i\} \xrightarrow{\text{a.s.}} 0$ .
- **Borel-Cantelli I:**  $\sum_n P(A_n) < \infty \Rightarrow P(A_n \text{ i.o.}) = 0$ ;
- **Borel-Cantelli II:** independent  $A_n$ ,  $\sum_n P(A_n) = \infty \Rightarrow P(A_n \text{ i.o.}) = 1$ .
- **a.s. Criterion:**  $X_n \xrightarrow{\text{a.s.}} X \Leftrightarrow \forall \epsilon > 0, P(|X_n - X| > \epsilon \text{ i.o.}) = 0$ .
  - Equiv.:  $\lim_{n \rightarrow \infty} P(\sup_{k \geq n} |X_k - X| > \epsilon) = 0, \forall \epsilon > 0$ ,
  - Enough:  $\forall m \geq 1, \sum_n P(|X_n - X| > 1/m) < \infty$ .
- **Subsequence:**  $X_n \xrightarrow{P} X \Rightarrow \exists n_k : X_{n_k} \xrightarrow{\text{a.s.}} X$ ;
  - proof: choose  $P(|X_{n_k} - X| > 1/k) < 2^{-k}$ .
- **Kolmogorov Ineq.:** independent RVs  $X_i$ ,  $E[X_i] = 0$ ,  $\text{var}(X_i) < \infty$ ,  $S_k = \sum_{i=1}^k X_i \Rightarrow P(\max_{k \leq n} |S_k| \geq a) \leq \text{var}(S_n)/a^2 = \sum_{i=1}^n \text{var}(X_i)/a^2$ .
- **Cantelli SLLN:** i.i.d.,  $E[X^4] < \infty \Rightarrow E[(\sum_{i=1}^n (X_i - \mu))^4] = O(n^2)$ ;  
Borel-Cantelli I  $\Rightarrow S_n/n \xrightarrow{\text{a.s.}} \mu$ .
- $\limsup A_n$  is event;  $\limsup P(A_n)$  is number;  
 $P(\limsup A_n) \geq \limsup P(A_n)$ .

Stochastic Processes

- **Stochastic Process:** A collection of RVs indexed by time.
  - **View as a mapping:**  $Z : T \times \Omega \rightarrow \mathbb{R}; (t, \omega) \mapsto Z_t(\omega)$ .
  - **View as a sequence:**  $\{Z_{t_1}, Z_{t_2}, \dots\}$ ,  $t_1, t_2 \in T$  is a sequence of time-relevant RVs defined on prob. space  $(\Omega, \mathcal{F}, P)$ .
  - Once the sample path  $\omega$  is fixed, values of the whole RV walk  $\{Z_{t_1}(\omega), Z_{t_2}(\omega), \dots\}$  are fixed.
- **Bernoulli Process:** discrete-time discrete-value SP  $\{X_i\}_{i \geq 1}$ ;  
 $X_i \sim$  i.i.d. Bernoulli( $p$ ).
  - **Intuition:** keep tossing a coin and record if  $i$ -th round is a head as  $X_i$ .
  - **Independence:** seq.  $X_{n+1}, X_{n+2}, \dots$  after  $n$  is a Bernoulli SP independent of  $X_1, \dots, X_n$ ,
  - **Memoryless:**  $W_n$  = trials from  $n$  to next success  $\sim \text{Geo}(p)$ ,

- **Interarrival:**  $Y_k$  = time of  $k$ -th success,  $T_1 = Y_1$ ,  $T_k = Y_k - Y_{k-1}$ ;  
 $T_i \sim$  i.i.d.  $\text{Geo}(p)$ ;  $Y_k = \sum_{i=1}^k T_i$ ,
- **$k$ -th success:**  $Y_k \sim \text{Pascal}(k, p)$ ,  
\*  $p_{Y_k}(t) = C_{t-1}^{k-1} p^k (1-p)^{t-k}$ ,  $t = k, k+1, \dots$ ,  
\*  $E[Y_k] = k/p$ ,  $\text{var}(Y_k) = k(1-p)/p^2$ ,
- **Splitting:** keep w.p.  $q \Rightarrow \text{Bernoulli SP}(pq)$ ,
- **Merging Independent Bernoullis:**  
Bernoulli( $p$ )+Bernoulli( $q$ )  $\Rightarrow$  Bernoulli( $p+q-pq$ ).
- **Poisson Process:** continuous-time discrete-value SP  $\{N_t\}_{t \geq 0}$ ,  $N_0 = 0$ ,  $N_t \sim \text{Poisson}(\lambda t)$  ;
  - **Intuition:** count the number of arrivals w.p.  $\lambda \delta$  for a small interval  $\delta$ , from 0 to  $t$  as  $N_t$ .
  - **Homogeneity:**  $P(k, t) = P(k \text{ arrivals in interval of duration } t)$   
 $= \sum_{n=0}^{\infty} P(N_{t+T} = n+k | N_T = n) P(n, T)$
  - **Independent:** Numbers of arrivals in **disjoint** time intervals are independent.
  - **Small  $\delta$ :**  $P(0, \delta) = 1 - \lambda \delta + o(\delta)$ ,  $P(1, \delta) = \lambda \delta + o(\delta)$ ,  
 $P(k, \delta) = o(\delta)$ ,  $k > 1$ ,
  - **Bernoulli Relation:** Select time infinitesimal  $\delta$ ,  $n = t/\delta$ ,  $p = \lambda \delta$ ;  
 $\{X_i\}$  is a Bernoulli SP( $p$ ),  $S_n = \sum_{k=1}^n X_i \sim \text{Binomial RV}(n, p) \rightarrow N_t \sim \text{Poisson RV}(\lambda t)$ ,  $n \rightarrow \infty$ ,
  - **First Arrival:**  $Y_1 = T_1 \sim \text{Exp}(\lambda)$ ;  $F_{Y_1}(t) = 1 - e^{-\lambda t}$ ,  $f_{Y_1}(t) = \lambda e^{-\lambda t}$ ;
  - **Interarrival:**  $Y_k$  = time of  $k$ th arrival,  $T_1 = Y_1$ ,  $T_k = Y_k - Y_{k-1}$ ;  
 $T_i \sim$  i.i.d.  $\text{Exp}(\lambda)$ ,  $Y_k = \sum_{i=1}^k T_i$ ,
  - **$k$ -th arrival:**  $Y_k \sim \text{Erlang}(\lambda, k)$ ,  
\*  $f_{Y_k}(y) = \lambda^k y^{k-1} e^{-\lambda y} / (k-1)!$ ,  $y \geq 0$ ;  
\*  $E[Y_k] = k/\lambda$ ,  $\text{var}(Y_k) = k/\lambda^2$ ,
  - **Splitting:** keep w.p.  $p \Rightarrow \text{Poisson}(\lambda p)$ .
  - **Merging Independent Poissons:**  $\text{Poisson}(\lambda_1) + \text{Poisson}(\lambda_2) = \text{Poisson}(\lambda_1 + \lambda_2)$ ;  
next arrival from SP 1 w.p.  $\lambda_1/(\lambda_1 + \lambda_2)$ .
- **Random Sum:**  $Y = \sum_{i=1}^N X_i$ ,  $Y = 0$  if  $N = 0$ ;  $\mathcal{M}_Y(s) = \mathcal{M}_N(\ln \mathcal{M}_X(s))$ .
- **Random Sum Table:**

|                          |                      |                            |
|--------------------------|----------------------|----------------------------|
| $X_i$                    | $N$                  | $Y$                        |
| Bernoulli( $p$ )         | Binomial( $m, q$ )   | Binomial( $m, pq$ )        |
| Bernoulli( $p$ )         | Poisson( $\lambda$ ) | Poisson( $\lambda p$ )     |
| Geo( $p$ )               | Geo( $q$ )           | Geo( $pq$ )                |
| Exponential( $\lambda$ ) | Geo( $q$ )           | Exponential( $\lambda q$ ) |

- Entropy
  - **Discrete:**  $H(X) = E[-\ln p_X(X)] = -\sum_k p_X(x_k) \ln p_X(x_k)$ .
  - **Continuous:**  $h(X) = E[-\ln f_X(X)] = -\int f_X(x) \ln f_X(x) dx$ .
  - **Bernoulli:**  $H(X) = -p \log_2 p - (1-p) \log_2 (1-p)$ ; max 1 at  $p = 1/2$ , min 0 at  $p = 0, 1$ .
  - **Max Entropy:**
    - finite  $\{x_1, \dots, x_n\} \Rightarrow p_k = 1/n$ ,
    - $[a, b] \Rightarrow U(a, b)$ ,
    - $[0, \infty)$ ,  $E[X] = 1/\lambda \Rightarrow \text{Exp}(\lambda)$ ,
    - $\mathbb{R}$ , fixed  $\mu$ ,  $\sigma^2 \Rightarrow N(\mu, \sigma^2)$ .

- Integration Formula
  - $\int x e^{ax} dx = \frac{1}{a^2} (ax - 1) e^{ax}$ ,  $\int x^2 e^{ax} dx = \frac{a^2 x^2 - 2ax + 2}{a^3} e^{ax}$ ,  $\int x^3 e^{ax} dx = \frac{a^3 x^3 - 3a^2 x^2 + 6ax - 6}{a^4} e^{ax}$ ,  $\int x^n e^{ax} dx = \frac{1}{a} x^n e^{ax} - \frac{n}{a} \int x^{n-1} e^{ax} dx$ .
  - $\int_0^{\infty} x^n e^{-ax} dx = n!/a^{n+1}$ .
  - $\int \sqrt{a^2 - x^2} dx = \frac{x}{2} \sqrt{a^2 - x^2} + \frac{a^2}{2} \arcsin \frac{x}{a}$ .
  - $\int \ln x dx = x \ln x - x$ ,  $\int x^n \ln x dx = \frac{1}{n+1} x^{n+1} \left( \ln x - \frac{1}{n+1} \right)$ .
  - $\int x \exp(-x^2/\sigma^2) dx = -\frac{\sigma^2}{2} \exp(-x^2/\sigma^2) + C$ .
  - $\int_{-\infty}^{+\infty} \exp(-(x-\lambda)^2/\mu^2) dx = \mu\sqrt{\pi}$ .
  - $\int_{-\infty}^{+\infty} x^2 \exp(-x^2/\mu^2) dx = \frac{\mu^3}{2} \sqrt{\pi}$ .